

# From Concept Generation to Concept Selection

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# Where We Are

## The Design Journey

| Week | Focus                        | Status         |
|------|------------------------------|----------------|
| 4    | Observations → Opportunities | ✓ Done         |
| 5    | Needs → Specifications       | ✓ Done         |
| 6    | <b>Concepts → Decision</b>   | 👉 Today        |
| 7    | Mockup                       | ➡<br>SOON Next |

# Today's Goal

## One clear concept, ready to build

By the end of today:

-  You have broken your problem into sub-functions
-  You have generated 3+ distinct concepts
-  You have selected ONE concept
-  You know what to prototype next week

# The Two Modes of Design Thinking

Today you will do both

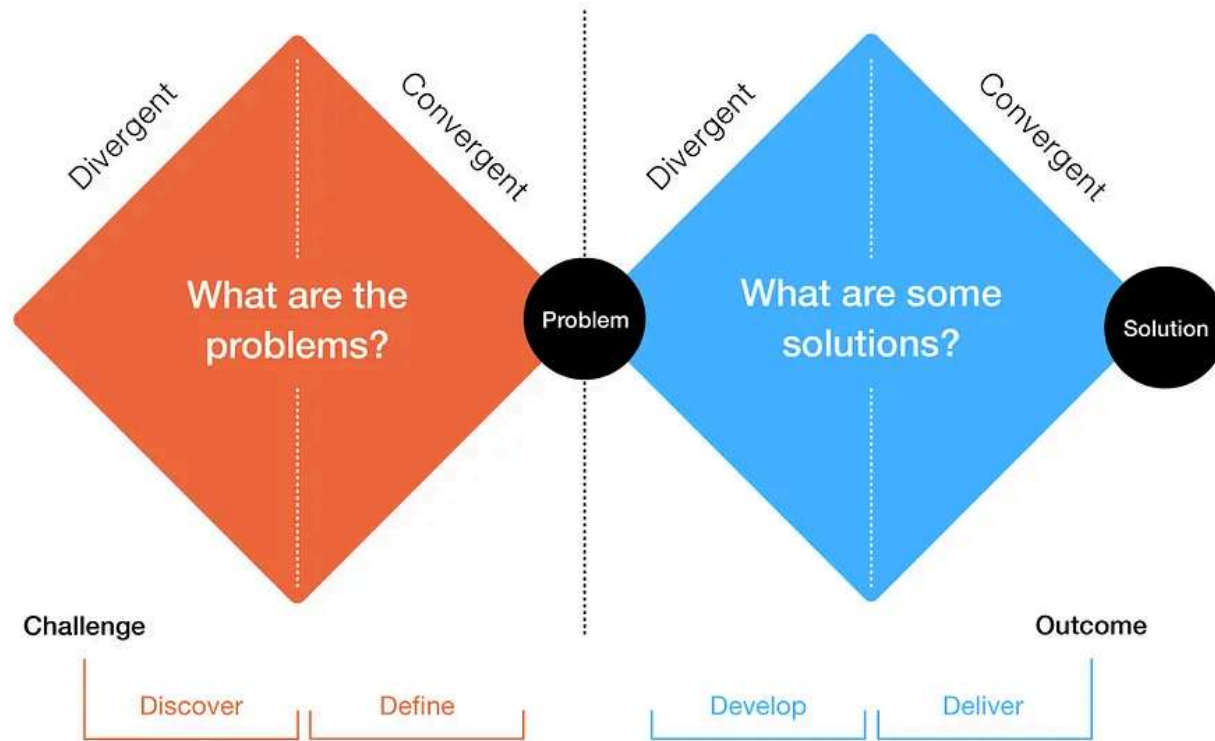
## Diverge first

- Generate many ideas
- No judgment
- Quantity over quality

## Then converge

- Evaluate systematically
- Make a justified decision
- Commit to one direction

## The Two Modes of Design Thinking (cont.)



Concept Generation → Concept Selection

# Why Not Just Pick the First Idea?

## The "First Idea Trap"

"We already know what we want to build."

✗ First ideas are usually:

- The most obvious
- The most familiar
- Not necessarily the best

☞ Structured generation forces you to explore the **solution space**

# Part 1: Concept Generation

## The 5-Step Process (Ulrich & Eppinger)

1. Clarify the problem
2. Search externally
3. Search internally
4. Explore systematically
5. Reflect on results

# Step 1: Clarify the Problem

## Functional Decomposition

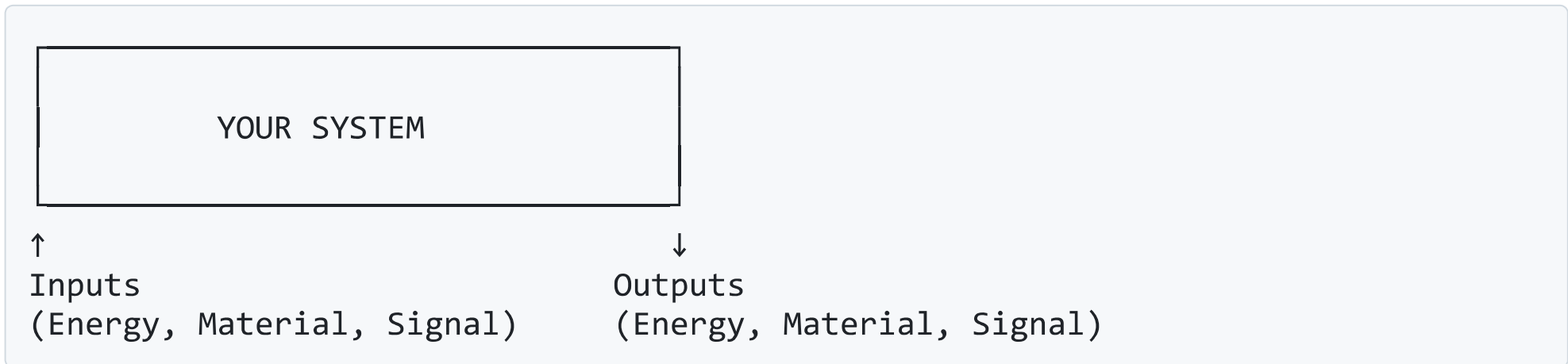
Break your system into:

**Inputs → Functions → Outputs**



# The Black Box Model

## Start simple



# Decompose Into Sub-Functions

## Break the black box open

Input → [Sense] → [Process] → [Act] → Output

↑↓

[Store]

↑↓

[Communicate]

# Example: Greenhouse Data Logger

## Sub-functions:

| Sub-function | What it does                    |
|--------------|---------------------------------|
| Sense        | Detect environmental conditions |
| Process      | Interpret and filter data       |
| Store        | Save readings locally           |
| Communicate  | Send data to user               |
| Power        | Supply energy to system         |
| Interface    | Allow user interaction          |

# Activity 1: Functional Decomposition

## Your Task:

1. Draw the **Black Box** for your system
2. Identify **4–6 sub-functions**
3. Label inputs and outputs

## Output:

- A diagram (hand-drawn is ok)
- List of sub-functions

## Step 2: External Search

### Where to look for inspiration

- **Lead users:** Experts who already solved similar problems
- **Patents:** Your Week 4 research (reuse it! You can find very valuable information in patents)
- **Analogous domains:** How does a different industry solve this?

# Analogous Thinking

## Examples for ICT Robotics:

| Your Problem                    | Analogous Domain        | Insight                      |
|---------------------------------|-------------------------|------------------------------|
| Detecting plant stress          | Medical diagnostics     | Non-invasive sensing         |
| Routing a robot in a greenhouse | Warehouse logistics     | Path planning algorithms     |
| Alerting workers                | Aviation cockpit alerts | Priority-based notifications |

👉 The best ideas often come from **outside your domain**

## Step 3: Internal Search

### Structured Brainstorming

⚠ **Common mistake:**

"Let's discuss as a group and pick the best idea."

✗ This leads to **groupthink**

# Individual First, Then Group

## Why?

- Individuals generate **more diverse ideas**
- Groups tend to **anchor on the first idea**
- Combining individual ideas = richer pool

## Method: **6-3-5 Brainwriting**

- 6 people, 3 ideas each, 5 minutes (your group is smaller, but the approach can be similar)
- Pass sheet → build on others' ideas



## Step 4: Systematic Exploration

### The Morphological Matrix

A powerful tool for concept generation

# Morphological Matrix

## Structure:

| Sub-function | Option A        | Option B   | Option C     |
|--------------|-----------------|------------|--------------|
| Sense        | Humidity sensor | Camera     | Manual input |
| Process      | Microcontroller | Smartphone | Cloud        |
| Communicate  | WiFi            | Bluetooth  | Display only |
| Power        | Battery         | Solar      | Wired        |
| Interface    | Button          | Voice      | Touchscreen  |

## Combining Into Concepts

**Draw "pathways" through the matrix**

**Concept 1 (Simple / Low-cost):**

Humidity sensor → Microcontroller → Display only → Battery → Button

**Concept 2 (High-tech / Automated):**

Camera → Cloud → WiFi → Solar → Voice

**Concept 3 (Hybrid / Creative):**

Manual input → Smartphone → Bluetooth → Battery → Touchscreen

## Activity 2: Morphological Matrix

### Your Task:

1. List your **sub-functions** (from Activity 1)
2. For each sub-function, generate **3+ solution options**
3. Draw **3 distinct pathways** (concepts)

### Rules:

-  Each concept must be genuinely different
-  At least one must be low-cost/simple
-  At least one must be high-tech/fully automated

# Quick Concept Sketches

## Visualize each concept

For each of your 3 concepts:

- Block diagram or sketch
- Label key components
- Note key interactions

👉 No CAD, no code — just **communicate the idea**

## Part 2: Concept Selection

### Now we converge

You have 3+ concepts.

👉 You need **one** by end of this class/week.

# Why Systematic Selection?

The "Gut Feeling" Problem: "We all like Concept B, let's go with that."

## ✗ Gut feeling:

- Ignores trade-offs
- Biased toward familiar tech
- Hard to justify to stakeholders

## ✓ Systematic selection:

- Transparent
- Linked to user needs
- Defensible

# Two-Step Selection Process

## Step 1: Screening (Pugh Matrix)

- Quick pass/fail filter
- Eliminates weak concepts

## Step 2: Scoring (Weighted Matrix)

- Detailed comparison
- Produces a ranked result



# Step 1: Pugh Concept Screening

## How it works:

- Choose a **reference concept** (usually the simplest)
- Compare others against it:
  - + = Better than reference
  - 0 = Same as reference
  - – = Worse than reference

## Pugh Matrix Example

| Criteria              | Reference (A) | Concept B | Concept C |
|-----------------------|---------------|-----------|-----------|
| Meets key needs       | REF           | +         | 0         |
| Technical feasibility | REF           | 0         | +         |
| Cost                  | REF           | –         | +         |
| Ease of use           | REF           | +         | –         |
| Score                 | 0             | +1        | +1        |

👉 Eliminate concepts with mostly –

## Step 2: Weighted Scoring Matrix

### Add importance weights (from Week 5)

| Criteria        | Weight | Concept A | Concept B | Concept C |
|-----------------|--------|-----------|-----------|-----------|
| Meets key needs | 5      | 3         | 4         | 5         |
| Feasibility     | 4      | 5         | 3         | 4         |
| Cost            | 3      | 4         | 5         | 2         |
| Ease of use     | 4      | 3         | 4         | 3         |
| <b>Total</b>    |        | <b>56</b> | <b>62</b> | <b>58</b> |

Weighted score =  $\Sigma$  (Weight  $\times$  Rating)

## Activity 3: Concept Selection

### Step 1: Pugh Screening

- Use your Week 5 needs as criteria
- Eliminate at least 1 concept

### Step 2: Weighted Scoring

- Use importance ratings from Week 5
- Rank remaining concepts

### Output:

- Completed matrices and **one selected concept**

# Making the Decision

## After scoring:

Ask yourselves:

- Does the winner **feel right**?
- If not — why? What did the matrix miss?
- Can you **combine** the best of two concepts?

👉 The matrix informs. **You** decide

# Documenting Your Decision

**For each concept, record:**

- Why it was **kept or eliminated**
- What **uncertainties** remain
- What **assumptions** you are making

👉 This is your design rationale — you will need it in Week 8 (Presentation)

# Part 3: Prototype Planning

## Bridge to Week 7

You have one concept.

👉 Now: what do you build?

# You Are NOT Building the Full System

## Week 7 Scope:

- ✗ Not a complete working robot
- ✗ Not a fully integrated system
- ✗ Not a polished product

✓ Physical model (mockup)

- Examples at [M3 Design](#), [Maker Pro](#), [Ruturaj](#), [Ritika](#), [Starfish Medical](#), [EPD](#), [Quirky](#), [Cor-mac](#), [Cal Maritime Makerspace](#), [Ponoko](#), [Helix Design](#), [Myszyński Design](#)

✓ One functional subsystem



# How to Choose What to Prototype

## Ask:

"What is the biggest risk in our concept?"

| Risk Type   | Example                       | Prototype Focus   |
|-------------|-------------------------------|-------------------|
| Technical   | "Can the sensor detect this?" | Sensing subsystem |
| Usability   | "Can workers use this?"       | Interface mockup  |
| Feasibility | "Can we build this cheaply?"  | Cost prototype    |

## Subsystem Examples (ICT Robotics)

| Concept Type      | Suggested Subsystem           |
|-------------------|-------------------------------|
| Monitoring system | Sensor + data display         |
| Worker interface  | Input device + feedback       |
| Automated robot   | Movement + obstacle detection |
| Data logging      | Sensor + local storage        |

## Activity 4: Prototype Planning

### Your Task:

1. Identify the biggest risk in your selected concept
2. Define one subsystem to prototype
3. List what you need:
  - Components, Tools, Skills

### Output:

- Subsystem definition
- Week 7 plan (1 paragraph)

# Assignment

## Concept Generation & Selection Report

In Moodle/Learn

## Key Takeaways

- Diverge before you converge
- Quantity first, quality later
- Selection must be systematic
- Prototype the **risk**, not the full system

Explore widely and decide deliberately

## Next Week: Mockup

You will:

- Build your chosen subsystem
- Test it against your specifications
- Evaluate using Design for X criteria
- Prepare for final presentation (Week 8)

## Suggested Reading

- Ulrich & Eppinger, Chapters on Concept Generation and Concept Selection  
(Chapters 6 and 7 in the 6th edition)